

Communication Systems

Computer Assignment 3



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1 Pulse Shaping and Baseband Signal Generation

In this section, we investigate the effect of pulse shaping on the time-domain waveform and the frequency spectrum of a BPSK signal. Two pulse shapes are considered: Rectangular (NRZ) and Half-Sine.

1.1 Waveform Visualization

The time-domain waveforms for 10 consecutive bits were generated. The rectangular pulse shows sharp, instantaneous transitions between symbols, while the half-sine pulse exhibits smoother transitions, ensuring continuity in the signal.

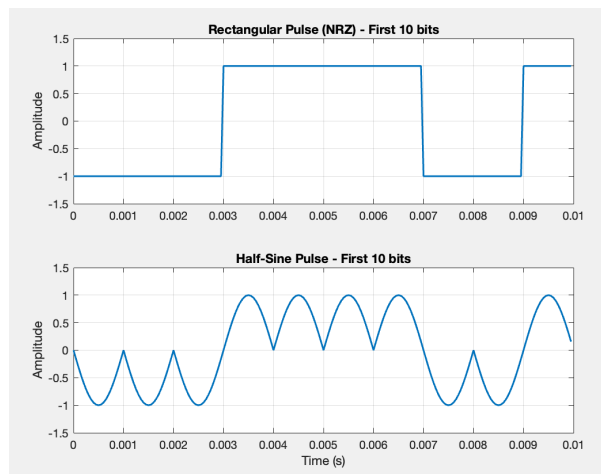


Figure 1: Time domain representation of BPSK signals with Rectangular and Half-Sine pulses.

1.2 Spectrum Estimation and Analysis

Using the Welch method (`pwelch`), the Power Spectral Density (PSD) for both signals was estimated.

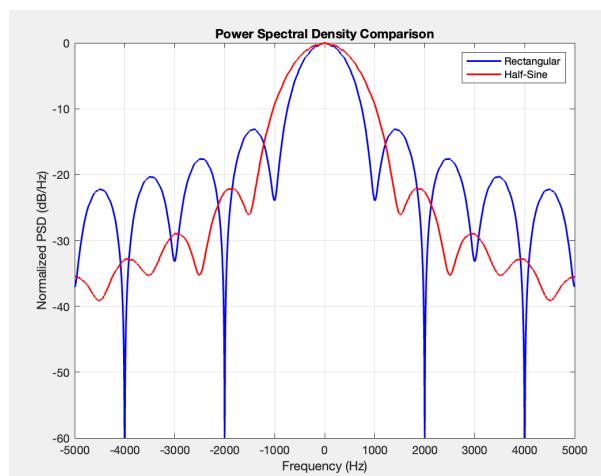


Figure 2: Power Spectral Density comparison.

Analysis:

- **Rectangular Pulse:** It has a narrower main lobe, indicating better spectral efficiency in the main band. However, its side lobes are high and decay slowly (proportional to $1/f^2$ in power). This causes significant Out-of-Band (OOB) radiation.
- **Half-Sine Pulse:** The main lobe is slightly wider (approx. $1.5\times$ wider than rectangular), but the side lobes decay much faster (proportional to $1/f^4$). This rapid decay is due to the smoother time-domain transitions, making it more suitable for channels with strict adjacent channel interference constraints.

2 Discrete-Time AWGN Modeling

2.1 Calibration Formula

To simulate the Additive White Gaussian Noise (AWGN) channel correctly in discrete time, the noise variance σ^2 must be calibrated based on the sampling frequency f_s and the desired E_b/N_0 . The relationship used in the code is:

$$\sigma^2 = \frac{N_0}{2} \times f_s = \frac{E_{pulse}}{2 \cdot 10^{\frac{SNR_{dB}}{10}}} \quad (1)$$

Where $L = f_s/R_b$ accounts for the oversampling factor. The energy is distributed over L samples, so the noise power must be scaled to maintain the correct signal-to-noise ratio per bit.

2.2 Noise Sensitivity

The simulation was run at $E_b/N_0 = 100$ dB and 0 dB. As expected, at 100 dB, the signal is virtually noise-free. At 0 dB, the noise amplitude is comparable to the signal amplitude, causing significant distortion.

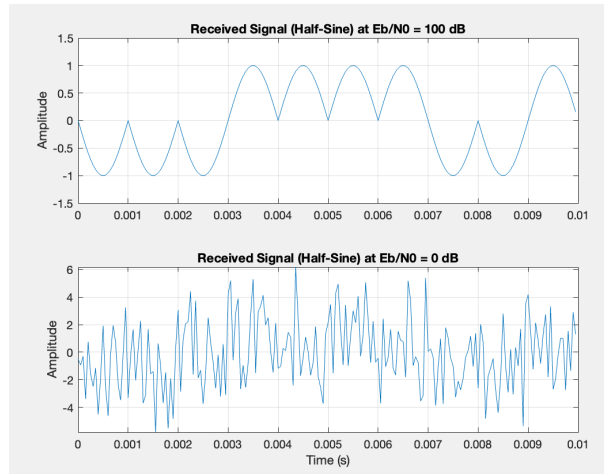


Figure 3: Effect of AWGN on the Half-Sine BPSK signal.

3 Matched Filter Receiver and SQNR Analysis

The optimal receiver in AWGN is the Matched Filter (MF), with impulse response $h[n] = p[L - 1 - n]$.

3.1 Matched Filter Output

The output of the MF for the half-sine pulse corresponds to the autocorrelation of the pulse shape. The optimal sampling points occur at the peaks (multiples of L), where the Signal-to-Noise Ratio (SNR) is maximized.

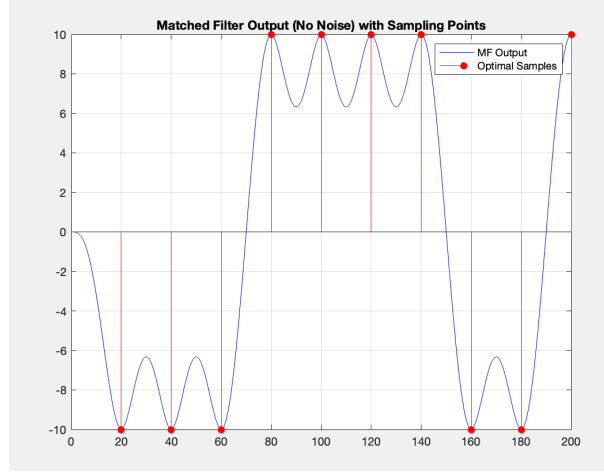


Figure 4: Matched Filter output showing optimal sampling instants.

3.2 SQNR vs. E_b/N_0

The Signal-to-Quantization/Noise Ratio (SQNR) at the output of the sampler was plotted against the input E_b/N_0 .

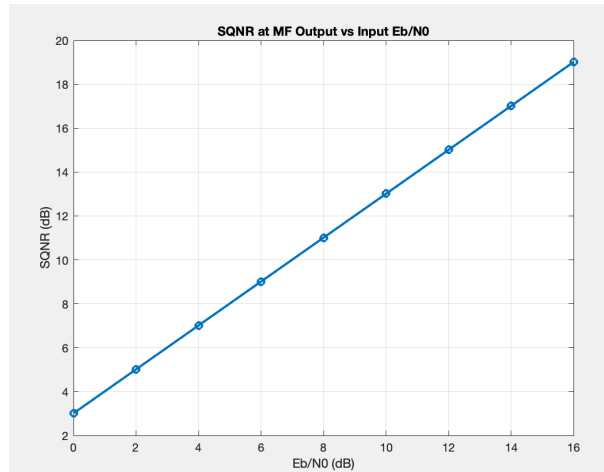


Figure 5: Linear relationship between Output SQNR and Input E_b/N_0 .

Linearity Analysis: Based on the simulation results, the slope of the curve was calculated as:

$$\text{Slope} \approx 1.0000$$

This confirms the linear relationship (in dB scale). Since the Matched Filter is a linear system, an increase of 1 dB in input signal power results directly in a 1 dB increase in output SNR. Ideally, the slope should be exactly 1.

4 BER Comparison

4.1 BER Curves

The Bit Error Rate (BER) was simulated for BPSK, QPSK, and 16-QAM using Monte Carlo simulations and compared with theoretical curves.

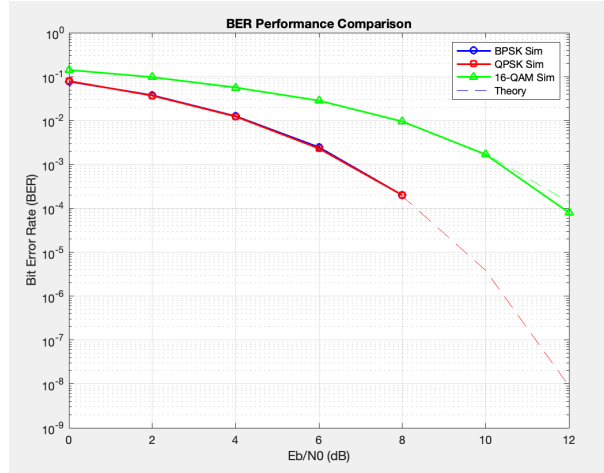


Figure 6: BER vs E_b/N_0 for BPSK, QPSK, and 16-QAM.

4.2 Numerical Analysis

The Mean Squared Error (MSE) between the simulated and theoretical BER values was computed to validate the simulation:

- **BPSK MSE:** 1.26×10^{-7}
- **QPSK MSE:** 1.22×10^{-7}
- **16-QAM MSE:** 9.20×10^{-7}

The very low MSE values indicate a highly accurate simulation.

4.3 Power Penalty Analysis

To achieve a target BER of 10^{-3} , the required E_b/N_0 values were found to be:

- **BPSK:** 6.72 dB
- **16-QAM:** 10.34 dB

Power Penalty: $10.34 - 6.72 = 3.63$ dB.

Interpretation: 16-QAM transmits 4 bits per symbol, packing constellation points closer together compared to BPSK. To maintain the same error probability, the energy per bit must be increased by approximately 3.6-4 dB to compensate for the reduced Euclidean distance between symbols.

5 Non-Ideal Effects

5.1 Phase Offset (15°)

A phase offset rotates the entire constellation. For QPSK, a 15° rotation reduces the noise margin but does not close the eye completely. However, at higher noise levels, this rotation increases symbol errors.

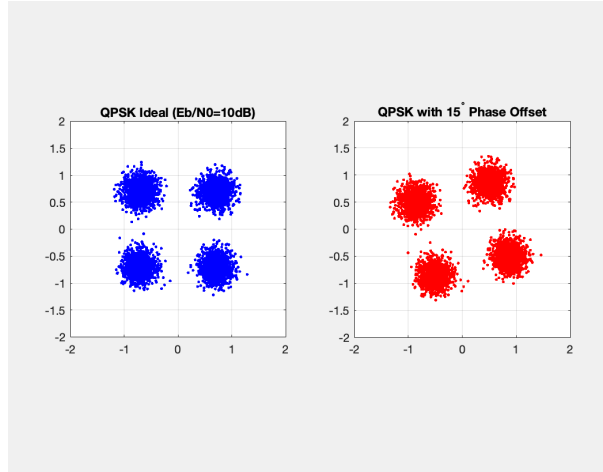


Figure 7: QPSK constellation before and after 15° phase shift.

5.2 Frequency Offset

A frequency offset Δf introduces a time-varying phase shift $\theta(t) = 2\pi\Delta f t$. This causes the constellation points to rotate continuously, forming a "ring" or circle instead of discrete points.

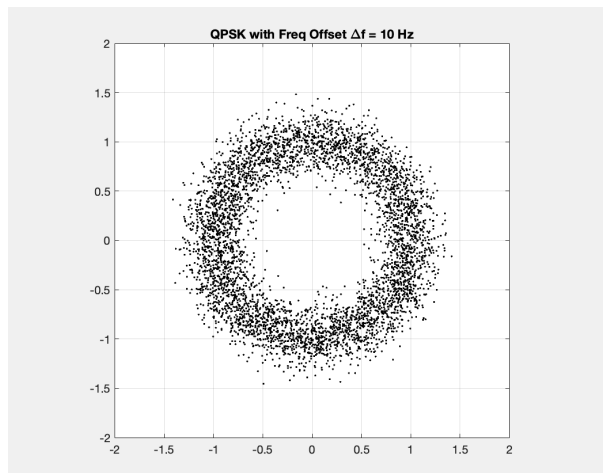


Figure 8: Constellation forming a ring due to frequency offset.

5.3 Eye Diagram

The Eye Diagram visualizes the signal quality.

- **Vertical Opening:** Indicates immunity to additive noise. A wider opening means better SNR.
- **Horizontal Opening:** Indicates immunity to timing jitter and ISI.

At 20 dB, the eye was observed to be "Wide Open," indicating minimal noise and ISI.

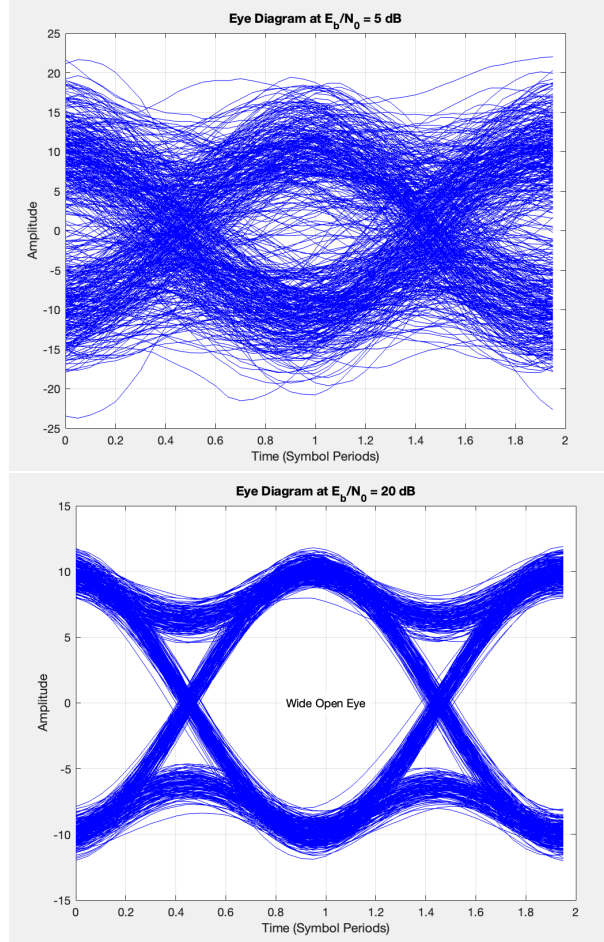


Figure 9: Eye Diagram at high SNR (20 dB).

6 System Frequency Response Analysis (Analog)

The analog receiver front-end consists of an RC high-pass filter and an ideal LPF. The total transfer function squared magnitude is:

$$|V(f)|^2 = |H_{RC}(f)|^2 \cdot |H_{LPF}(f)|^2 = \frac{(2\pi fRC)^2}{1 + (2\pi fRC)^2} \cdot \text{rect}\left(\frac{f}{2W}\right) \quad (2)$$

Based on the simulation parameters ($R = 1, C = 1$), the RC corner frequency is calculated as $f_{3dB} \approx 0.1592$ Hz. With the ideal LPF cutoff at $W = 5$ Hz, the effective passband of the system is approximately **0.1592 Hz to 5.0000 Hz**.

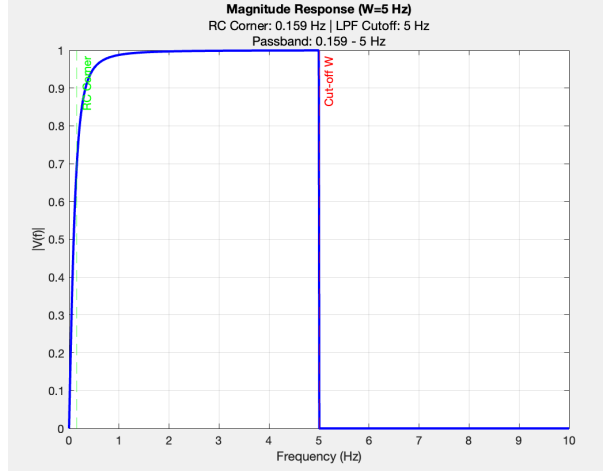


Figure 10: Magnitude response $|V(f)|$ for $W = 5$ Hz.

7 SNR Derivation and Functional Plotting

7.1 Output SNR Plot

The signal power (P_s) and noise power (P_n) at the output were derived analytically. The theoretical relationship for output SNR as a function of bandwidth W is given by:

$$SNR(W) = \frac{A_c^2}{N_0} \frac{\frac{(2\pi f_c RC)^2}{1 + (2\pi f_c RC)^2}}{W - \frac{1}{2\pi RC} \arctan(2\pi RCW)} \quad (3)$$

Using this relationship, the SNR was plotted for varying W (where $W > f_c$).

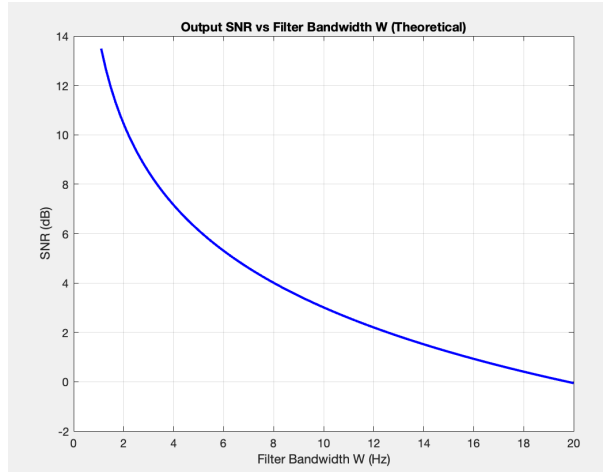


Figure 11: Output SNR as a function of LPF bandwidth W .

7.2 Discussion on SNR Behavior

As observed in the plot above:

- **Peak SNR:** The SNR is maximized around $W \approx f_c$. This occurs because at this point, the filter bandwidth is just wide enough to pass the signal frequency f_c completely, while minimizing the amount of noise allowed into the system.

- **Decreasing Trend:** For $W > f_c$, the SNR starts to decrease. This happens because the signal power remains constant (since the signal at f_c is already fully within the passband), but the noise power continues to increase as the bandwidth W widens, admitting more high-frequency noise components.
- **Optimal Bandwidth:** For a fixed f_c , the optimal bandwidth is slightly above f_c (practically $W \approx f_c$) to capture the signal while rejecting maximum noise.

8 Effect of Cut-off Frequency Mismatch

We evaluated the case where the LPF bandwidth exactly matches the signal frequency ($W = f_c = 1$ Hz).

- **SNR at $W = f_c$:** 14.01 dB.
- **Signal Power Loss:** Since f_c is included in the passband $W = f_c$, ideally there is **0.00 dB** loss compared to $W \rightarrow \infty$ regarding the signal component itself. The filter allows the fundamental frequency to pass.

The output noise PSD for $W = 10$ Hz shows the shaping effect: low frequencies are attenuated by the RC filter (High-Pass), and frequencies above 10 Hz are cut off by the LPF.

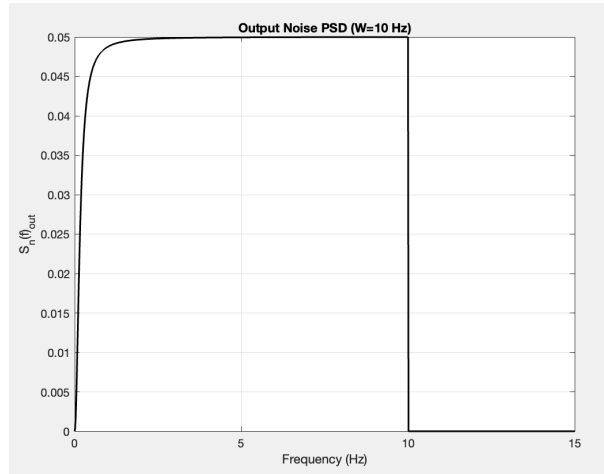


Figure 12: Power Spectral Density of the noise at the receiver output.

9 Monte Carlo Verification

A time-domain Monte Carlo simulation was performed to verify the theoretical SNR derivation. The simulation averaged the power of signal and noise over multiple realizations.

Simulation Results:

- $W = 1.5$ Hz $\rightarrow$ SNR = 11.78 dB
- $W = 3.0$ Hz $\rightarrow$ SNR = 8.49 dB

- $W = 5.0 \text{ Hz} \rightarrow \text{SNR} = 6.06 \text{ dB}$
- $W = 10.0 \text{ Hz} \rightarrow \text{SNR} = 2.97 \text{ dB}$
- $W = 15.0 \text{ Hz} \rightarrow \text{SNR} = 1.20 \text{ dB}$

As shown in the figure below, the Monte Carlo simulation points (red dots) perfectly align with the theoretical curve, validating the derived equations.

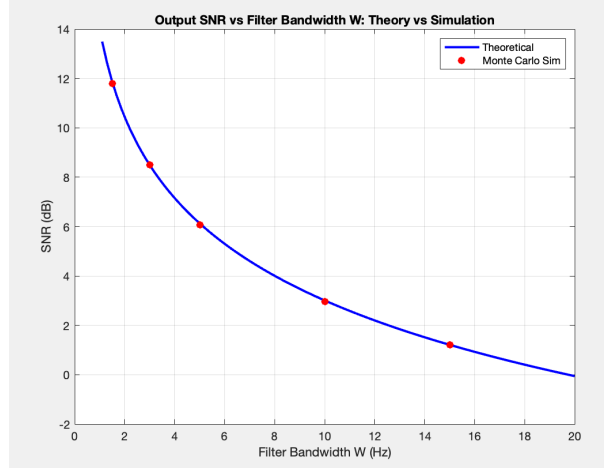


Figure 13: Verification of theoretical SNR using Monte Carlo simulation.

10 Bonus: Equivalent Noise Bandwidth

10.1 Mathematical Derivation

The Equivalent Noise Bandwidth (B_{eq}) is defined as the bandwidth of an ideal rectangular filter that passes the same total noise power as the actual system, assuming the peak gain of both filters is the same. The mathematical definition is:

$$B_{eq} = \frac{\int_0^\infty |H(f)|^2 df}{|H(f)|_{max}^2} \quad (4)$$

For our system, the magnitude response squared is:

$$|H(f)|^2 = \frac{(2\pi fRC)^2}{1 + (2\pi fRC)^2}, \quad \text{for } 0 \leq f \leq W \quad (5)$$

Since $|H(f)|^2$ is strictly increasing with frequency in the passband, the maximum gain occurs at the cutoff frequency $f = W$:

$$|H(f)|_{max}^2 = |H(W)|^2 = \frac{(2\pi WRC)^2}{1 + (2\pi WRC)^2} \quad (6)$$

The total noise power integral (numerator) is calculated as:

$$P_{total} = \int_0^W \frac{(2\pi fRC)^2}{1 + (2\pi fRC)^2} df = W - \frac{1}{2\pi RC} \arctan(2\pi RCW) \quad (7)$$

Substituting these into the definition of B_{eq} :

$$B_{eq} = \frac{W - \frac{1}{2\pi RC} \arctan(2\pi RCW)}{\frac{(2\pi WRC)^2}{1+(2\pi WRC)^2}} \quad (8)$$

Which simplifies to:

$$B_{eq} = \left(1 + \frac{1}{(2\pi WRC)^2}\right) \left(W - \frac{1}{2\pi RC} \arctan(2\pi RCW)\right) \quad (9)$$

10.2 Calculation

Using the simulation parameters $R = 1$, $C = 1$, and $W = 5$ Hz:

$$B_{eq} \approx 4.7551 \text{ Hz}$$

10.3 Conceptual Question

If the RC High-Pass filter were replaced by a second-order Butterworth filter, the noise power at the output would **decrease**. A 2nd-order Butterworth filter has a steeper roll-off (40 dB/dec) compared to the RC filter (20 dB/dec). This sharper transition would reject more low-frequency noise components near DC, resulting in a lower total noise power integral and thus a higher SNR.